Exp-2

→ Algo:

1. if $n <= 2$ return n
2. for each point i:
   - create map slope → count
   - set same = 0, maxi = 0
3. for every $j > i$:
   - compute dx, dy
   - if duplicate → same ++
   - else:
     - reduce (dy, dx) using gcd
     - increment map count
     - update maxi.
4. ans = max(ans, maxi + same + 1)
5. return ans.

→ Code:

```
int maxPoints( vector< vector <int>> & points) {
    int n = points.size();
    if (n <= 2) return n;
    int ans = 0;
```

```
for(int i=0; i<n; i++){
    map<pair<int,int>, int> mp;
    int same=0, maxi=0;

    for(int j=i+1; j<n; j++){
        int x1 = points[i][0], y1 = points[i][1];
        int x2 = points[j][0], y2 = points[j][1];

        int dx = x2-x1;
        int dy = y2-y1;
        if(dx==0 && dy==0){
            same++;
            continue;
        }
        int g = __gcd(dx,dy);
        dx /= g;
        dy /= g;
        mp[{dy,dx}]++;
        maxi = max(maxi, mp[{dy,dx}]);

    }
    ans = max(ans, maxi+same+1);
} return ans;
}
```